9.5.10

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October 4, 2025

Question

If the sum of zeroes of the polynomial is $\sqrt{2}$, then find the value of k .

$$p(x) = 2x^2 - k\sqrt{2}x + 1$$

Solution

$$\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \tag{1}$$

$$\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}. \tag{2}$$

Substituting the line into the conic gives

$$(\mathbf{h} + \kappa \mathbf{m})^\top \mathbf{V}(\mathbf{h} + \kappa \mathbf{m}) + 2\mathbf{u}^\top (\mathbf{h} + \kappa \mathbf{m}) + f = 0 \tag{3}$$

$$\Rightarrow \kappa^2(\mathbf{m}^\top \mathbf{V}\mathbf{m}) + 2\kappa\, \mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) + (\mathbf{h}^\top \mathbf{V}\mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f) = 0 \tag{4}$$

$$\kappa_1 + \kappa_2 = -\frac{2\, \mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u})}{\mathbf{m}^\top \mathbf{V}\mathbf{m}}. \tag{5}$$

Given:(Roots are given by intersection of conic with x-axis)

$$\mathbf{V} = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -\frac{k\sqrt{2}}{2} \\ 0 \end{pmatrix}, \quad f = 1 \tag{6}$$

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{7}$$

Solution

$$\mathbf{Vh} + \mathbf{u} = \begin{pmatrix} -\frac{k\sqrt{2}}{2} \\ 0 \end{pmatrix}, \tag{9}$$

$$\mathbf{m}^\top \mathbf{Vm} = 2. \tag{10}$$

$$\mathbf{m}^\top (\mathbf{Vh} + \mathbf{u}) = -\frac{k\sqrt{2}}{2} \tag{11}$$

$$\kappa_1 + \kappa_2 = -\frac{2(-\frac{k\sqrt{2}}{2})}{2} = \sqrt{2}. \tag{12}$$

$$\Rightarrow \boxed{k = 2}. \tag{13}$$

C Code

```
        #include <stdio.h>
#include <math.h>

int main() {
    // Given conic: 2x^2 - k2 x + 1 = 0
    // Matrices and vectors as per conic representation:
    // x^T V x + 2u^T x + f = 0

    double V[2][2] = {{2, 0}, {0, 0}};
    double h[2] = {0, 0};
    double m[2] = {1, 0};
    double f = 1;
    double k; // unknown to be determined

    // Given condition: (sum of roots) = 2
    double given_sum = sqrt(2);

    // From formula:
```

C Code

```c
//  +  = -[2 * (m(Vh + u))] / (m V m)
// For given h = 0, m = [1 0] :
// m V m = 2, (Vh + u) = [-k2/2, 0]
//   +  = -[2 * (-k2/2)] / 2 = (k2)/2

// Equating to given sum 2  (k2)/2 = 2  k = 2

k = 2.0;

// Verification step:
double u[2] = { -k * sqrt(2) / 2.0, 0.0 };
double Vh_plus_u[2];
Vh_plus_u[0] = V[0][0]*h[0] + V[0][1]*h[1] + u[0];
Vh_plus_u[1] = V[1][0]*h[0] + V[1][1]*h[1] + u[1];

double mT_V_m = m[0]*(V[0][0]*m[0] + V[0][1]*m[1]) +
                m[1]*(V[1][0]*m[0] + V[1][1]*m[1]);
```

C Code

```c
    double mT_Vh_u = m[0]*Vh_plus_u[0] + m[1]*Vh_plus_u[1];

    double kappa_sum = -2 * mT_Vh_u / mT_V_m;

    printf(Computed  +  = %.4f\n, kappa_sum);
    printf(Given sum of roots = %.4f\n, given_sum);
    printf(Value of k = %.2f\n, k);

    return 0;
}
```

Python Direct

```
import numpy as np
import matplotlib.pyplot as plt

from libs.line.funcs import *
from libs.triangle.funcs import *
from libs.conics.funcs import *

# Given polynomial: p(x) = 2x - k2 x + 1

k = 2
V = np.array([[2, 0],
              [0, 0]])
u = np.array([[-k*np.sqrt(2)/2],
              [0]])
f = 1
```

Python Direct

```
# Intersection with x-axis: y = 0
# Substitute in conic: 2x - k2 x + 1 = 0

a = 2
b = -k*np.sqrt(2)
c = 1

# Use function from funcs to compute roots
x1, x2 = quad_roots(a, b, c)
x_vals = np.array([x1, x2])
y_vals = np.zeros(2)

# Compute sum of roots (verification)

sum_roots = x1 + x2
print(fSum of zeroes = {np.round(sum_roots,3)}  2)

# Plot the conic and x-axis intersection
```

Python Direct

```
x_plot = np.linspace(min(x_vals) - 1, max(x_vals) + 1, 400)
y_plot = 2*x_plot**2 - k*np.sqrt(2)*x_plot + 1

plt.figure(figsize=(7, 5))
plt.axhline(0, color='black', linewidth=1.2, label='x-axis')
plt.plot(x_plot, y_plot, 'b', label=r'$2x^2 - 2\sqrt{2}x + 1$')
plt.scatter(x_vals, y_vals, color='red', zorder=5)

# Label intersection points inside the plot
for i in range(2):
    label_point(x_vals[i], y_vals[i],
                f({np.round(x_vals[i],2)}, {np.round(y_vals[i],2)
                    }),
                offset=(0,0.2), color='darkred')
```

Python Direct

```
# Display sum of zeroes
plt.text(np.mean(x_vals), max(y_plot)/2,
         fSum of zeroes = {np.round(sum_roots, 3)}  2,
         fontsize=10, color='purple', ha='center')

plt.title(Intersection of Conic with x-axis)
plt.xlabel(x)
plt.ylabel(y)
plt.legend()
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Python Shared

```
        import numpy as np
import matplotlib.pyplot as plt
import ctypes

# Load shared library
lib = ctypes.CDLL(./conic_kappa.so)

# Define argument/return types
lib.kappa_sum.restype = ctypes.c_double
lib.kappa_sum.argtypes = [
    np.ctypeslib.ndpointer(dtype=np.float64, shape=(2,2)),
    np.ctypeslib.ndpointer(dtype=np.float64, shape=(2,)),
    np.ctypeslib.ndpointer(dtype=np.float64, shape=(2,)),
    np.ctypeslib.ndpointer(dtype=np.float64, shape=(2,))
]

lib.compute_k.restype = ctypes.c_double
```

Python Shared

```
# Given data (same as LaTeX)
# -----------------------------
k = 2.0
V = np.array([[2, 0],
              [0, 0]], dtype=np.float64)
u = np.array([-k*np.sqrt(2)/2, 0], dtype=np.float64)
h = np.array([0, 0], dtype=np.float64)
m = np.array([1, 0], dtype=np.float64)

# Call C function for kappa sum
kappa_sum_val = lib.kappa_sum(V, u, h, m)
print(Computed ( + ) from .so =, np.round(kappa_sum_val, 4))

# Check against 2
print(Expected =, np.round(np.sqrt(2), 4))

# Plot the conic and intersection points
a, b, c = 2, -k*np.sqrt(2), 1
D = b**2 - 4*a*c
```

Python Shared

```
x1 = (-b + np.sqrt(D)) / (2*a)
x2 = (-b - np.sqrt(D)) / (2*a)
x_vals = np.array([x1, x2])
y_vals = np.zeros(2)

# Generate parabola
x_plot = np.linspace(min(x_vals)-1, max(x_vals)+1, 400)
y_plot = 2*x_plot**2 - k*np.sqrt(2)*x_plot + 1

plt.figure(figsize=(7,5))
plt.axhline(0, color='black', linewidth=1)
plt.plot(x_plot, y_plot, 'b', label=r'$2x^2 - 2\sqrt{2}x + 1$')
plt.scatter(x_vals, y_vals, color='red')
```

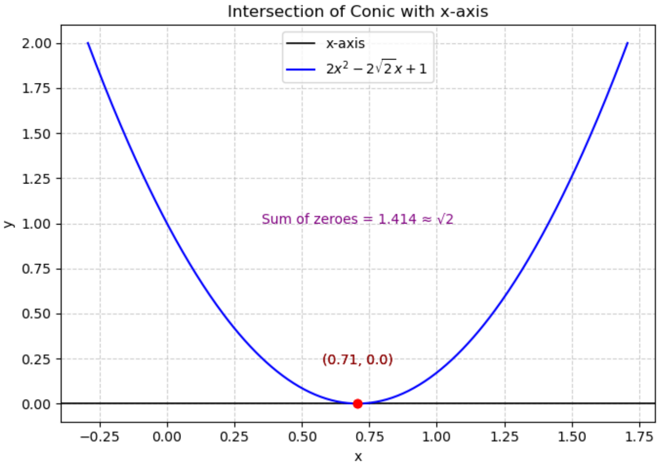
Python Shared

```
# Label points
for (x,y) in zip(x_vals, y_vals):
    plt.text(x, y+0.2, f({np.round(x,2)}, {np.round(y,2)}),
             ha='center', va='bottom', color='darkred')

plt.text(np.mean(x_vals), max(y_plot)/2,
         f(+) from C = {np.round(kappa_sum_val,3)}  2,
         fontsize=10, color='purple', ha='center')

plt.legend()
plt.title( +  from C shared library (Conic Intersection))
plt.xlabel(x)
plt.ylabel(y)
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Plot



Intersection of Conic with x-axis
x-axis
$2x^2 - 2\sqrt{2}x + 1$
Sum of zeroes = 1.414 ≈ √2
(0.71, 0.0)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
y
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
x

Figure: